

Genomics of schizophrenia, bipolar disorder and major depressive disorder

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Abstract

Schizophrenia, bipolar disorder and major depressive disorder – which are the most common adult disorders requiring psychiatric care – contribute substantially to premature mortality and morbidity globally. Treatments for these disorders are suboptimal, there are no diagnostic pathologies or biomarkers and their pathophysiologies are poorly understood. Novel therapeutic and diagnostic approaches are thus badly needed. Given the high heritability of psychiatric disorders, psychiatry has potentially much to gain from the application of genomics to identify molecular risk mechanisms and to improve diagnosis. Recent large-scale, genome-wide association studies and sequencing studies, together with advances in functional genomics, have begun to illuminate the genetic architectures of schizophrenia, bipolar disorder and major depressive disorder and to identify potential biological mechanisms. Genomic findings also point to the aetiological relationships between different diagnoses and to the relationships between adult psychiatric disorders and childhood neurodevelopmental conditions.

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Introduction

Psychiatric disorders are characterized by clinically important disturbances in thinking, emotional regulation and behaviour. Their global burden on individual health and societal welfare is considerable, and they are major causes of premature mortality and morbidity¹. Schizophrenia, bipolar disorder and major depressive disorder (MDD) are among the most severe, adult-onset psychiatric disorders in terms of their impact on the individual and society², and they have substantial overlap in clinical features (Box 1 and Table 1).

From the late eighteenth century and throughout the nineteenth century, physicians and psychiatrists regarded heredity as one of the strongest risk factors for 'insanity'³. The twentieth century saw the recognition of different syndromes in mental illness and the development of broadly agreed diagnostic criteria that form the basis of current classifications and clinical practice. It also saw the development and refinement of the key tools of classical genetic epidemiology – namely, family, twin and adoption studies. Psychiatric disorders were extensively studied using these methods, providing robust evidence for the heritability of the different diagnostic categories (Table 1).

Psychiatric genetics was thus placed on a sound empirical footing for the application of molecular genetic approaches. These offered the prospect of identifying specific risk alleles and genes, thereby providing routes into understanding biological mechanisms and the development of novel approaches for stratification and diagnosis, as well as biomarkers to aid diagnosis and treatment targeting. The use of molecular genetic approaches was particularly attractive in psychiatry as, given the complexity and inaccessibility of the brain, pathophysiology remains poorly

understood, and, in the absence of established disease mechanisms, available drug treatments are largely based on serendipitous discoveries.

Linkage and candidate gene approaches for psychiatric disorders beginning in the 1980s were largely unsuccessful. However, robust findings have steadily accumulated since the advent of genome-wide association studies (GWAS) and copy number variant (CNV) analysis and the development of next-generation sequencing methods, together with the formation of consortia assembling large and sufficiently powered samples⁴. Over the past 10 years, a massive collaborative effort in psychiatric genomics, culminating in recently published large GWAS and whole exome-sequencing studies, has identified many common and rare risk alleles and loci and has begun to reveal the genetic architectures of psychiatric disorders.

The focus of this Review Article is the genomics of schizophrenia, bipolar disorder and MDD, the adult-onset psychiatric disorders for which genomic approaches have so far provided the greatest insight. We do not cover neurodegenerative disorders such as Alzheimer's disease, as these are largely genetically and neuropathologically distinct from psychiatric disorders⁵, nor do we focus on childhood-onset neurodevelopmental conditions (NDCs) such as autism, attention deficit hyperactivity disorder (ADHD) and intellectual disability, although we do discuss the genetic relationship between NDCs and adult psychiatric disorders. We also briefly consider other psychiatric disorders in relation to genetic overlap (pleiotropy) with schizophrenia, bipolar disorder and MDD.

To provide context, we start by briefly summarizing the main features of these psychiatric disorders. We describe the current understanding of their genetic architectures and the extent to which genomic

Box 1 | Psychiatric diagnoses

In the absence of diagnostic tests and biomarkers, psychiatric diagnoses are based solely on clinical signs and symptoms. The need for diagnostic standardization and inter-rater reliability has been addressed by the operationalised classifications in the Diagnostic and Statistical Manual of Mental Disorders (DSM) and the International Classification of Diseases (ICD)¹⁵⁰. These classifications have had many benefits for clinical practice and research but were never intended to be valid descriptors of distinct disease entities sharing an underlying pathogenic process^{151,152}. This lack of discrete, biologically valid, diagnostic categories in psychiatry provides a plausible explanation for the pervasive pleiotropy seen between psychiatric diagnoses and is consistent with observations that the genetic liability for specific clinical features (such as depression, mania and psychosis) is at least partially shared, irrespective of the primary categorical diagnosis (see main text).

The clinical features of schizophrenia, bipolar disorder and major depressive disorder (MDD) substantially overlap (Table 1). Consequently, the boundaries between disorders are indistinct, as are the boundaries between these disorders and psychological phenomena present in people who are not classified as being unwell¹⁵⁰. Moreover, individuals with the same diagnosis vary widely in disease symptoms, severity, course and outcome, which suggests that there is mechanistic heterogeneity within diagnostic groups¹⁵⁰. The lack of clear boundaries between diagnoses and the heterogeneity within them are reflected in the common assignment of diagnoses outside of the three main groupings of schizophrenia,

bipolar disorder and MDD. For example, schizoaffective disorder is a diagnosis given to individuals who have the psychotic features of schizophrenia as well as prominent symptoms of mood disorder. This is sometimes divided into bipolar and depressive subtypes depending on whether manic episodes have occurred. Genetic epidemiology suggests that schizoaffective disorder shares genetic risk with both schizophrenia and bipolar disorder and may represent an intermediate form in a schizophrenia–bipolar disorder continuum²⁶. Moreover, two types of bipolar disorder are recognized. Bipolar disorder type 1 is characterized by the presence of mania and, usually, episodes of depression, whereas bipolar disorder type 2 is characterized by hypomania and episodes of depression.

Early genomic studies used the Diagnostic and Statistical Manual of Mental Disorders or International Classification of Diseases definitions of schizophrenia, bipolar disorder and MDD derived from structured interviews. Subsequently, resulting from the need for larger samples, researchers have increasingly used less stringent approaches, potentially resulting in the inclusion of cases who meet the diagnostic criteria for another disorder or who have relatively mild symptoms and do not meet the criteria for any disorder³⁴. Such studies also lack sufficiently granular clinical data for the investigation of heterogeneity and transdiagnostic syndromes and symptoms. This has led to calls for new cohorts for genomic research in psychiatry to be established, with deeper and more diverse phenotypes^{34,153}.

Table 1 | Main features of schizophrenia, bipolar disorder and MDD

Feature	Schizophrenia	Bipolar disorder	MDD
Core symptoms	Delusions, hallucinations, formal thought disorder and negative symptoms	Episodes of mania (BD1) or hypomania (BD2), together with episodes of depression	Episodes of depression
Additional symptoms	Cognitive impairment and episodes of mania and depression	Delusions and hallucinations, cognitive impairment	Delusions and hallucinations, cognitive impairment
Estimated global prevalence	24 million cases ¹³⁹	40 million cases ¹³⁹	280 million cases ¹³⁹
Usual age at diagnosis	16–30 years ¹⁴⁰	15–30 years ¹⁴¹	25–35 years ¹⁴²
Commonly used drug treatments	Antipsychotics, antidepressants and mood stabilizers	Mood stabilizers, antipsychotics	Antidepressants, antipsychotics, mood stabilizers
Heritability estimates from twin studies	0.64–0.81 (refs. 24,143)	0.59–0.85 (refs. 24,27)	0.32–0.75 (refs. 24,27,144)
Genetic liability from RCVs	0.02 (ref. 7)	0.02 (ref. 145)	0.007 (extrapolated from ref. 52)
Genetic liability from CNVs	0.02 (ref. 7)	Unknown	Unknown
SNP heritability ^a	0.25 (ref. 7)	0.21 (ref. 36)	0.08 (ref. 33)
SNP heritability from GWS variants ^a	0.03 (ref. 7)	0.04 (ref. 36)	0.01 (ref. 33)

BD1, bipolar disorder type 1; BD2, bipolar disorder type 2; CNVs, copy number variants; GWAS, genome-wide association studies; GWS, genome-wide significant; MDD, major depressive disorder; RCVs, rare coding variants; SNP, single-nucleotide polymorphism. ^aSNP heritability estimates are derived from European populations, which currently are over-represented in GWAS. Note that estimates from rare variants include contributions from transmitted and de novo mutations, although the latter do not contribute to heritability.

findings account for heritability estimates from genetic epidemiology. We consider the biological insights that have emerged from genomic studies, including the brain regions and cell types that are likely to mediate genetic risk. We discuss the evidence for genetic relationships across psychiatric disorders and how this informs our understanding of the aetiological relationships between different diagnostic groups. We present evidence implicating genes, biological pathways, cell types and potential mechanisms that cross traditional diagnostic boundaries, and we consider the clinical and research implications of these findings as well as the challenges that lie ahead.

There have been recent reviews of genomic studies of individual psychiatric disorders^{6–9}, as well as of psychopathology more broadly^{10,11}. Here, by focusing on the three most prominent adult psychiatric conditions, we provide a concise summary of the most robust findings for each, including very recently published, large-scale studies of bipolar disorder and MDD. We emphasize evidence for the genetic similarities and differences between schizophrenia, bipolar disorder and MDD and how these conditions differ in terms of their genetic relationships with NDCs and the contribution of neurodevelopmental processes. We also highlight the implications of variations in diagnostic stringency on the outcomes of genomic studies and the need for more research that integrates deeper phenotyping and biomarkers with genomics research.

Overview of schizophrenia, bipolar disorder and depression

We start by providing a brief overview of the clinical features, pathophysiology and genetic epidemiology of schizophrenia, bipolar disorder and depression to provide context for the remainder of this Review Article.

Clinical features

Psychiatric diagnoses are syndromic, based solely on the presence or absence of constellations of signs and symptoms. There are no clinically validated diagnostic tests or biomarkers for these disorders (Box 1). The main clinical features of schizophrenia, bipolar disorder and MDD are psychotic symptoms (delusions and hallucinations), negative

symptoms (impaired motivation, reduction in spontaneous speech and social withdrawal), disturbances of mood and activity that can either be low (depression) or elevated (mania or hypomania) and impairment of cognition (Table 1). Although the relative prominence of each of these features varies across the three diagnoses, there is substantial clinical overlap (Box 1).

Pathophysiology

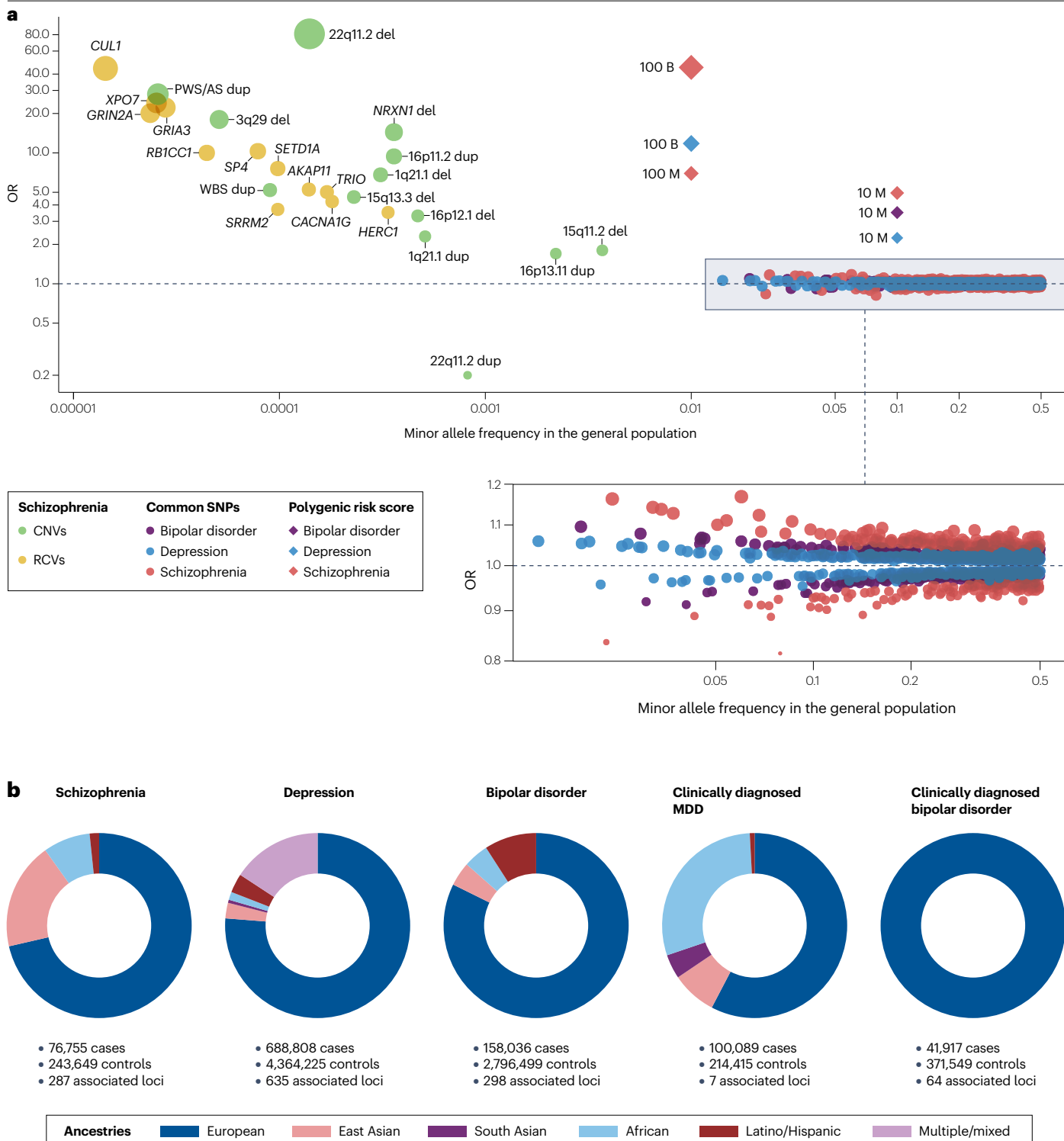
There is no diagnostic neuropathology for schizophrenia, bipolar disorder or MDD, and despite decades of neurobiological research and many hypotheses, their pathophysiology remains poorly understood – apart from evidence that psychosis results, at least in part, from dopamine dysfunction¹². There is evidence from various sources that schizophrenia partly reflects disturbances in brain development^{13,14} and that these might lie on a neurodevelopmental continuum, with a gradient of decreasing severity from congenital and childhood NDCs, through schizophrenia, to mood disorders (bipolar disorder and MDD)^{15,16}.

Brain imaging studies have not identified structural differences that permit individuals with one of the three disorders to be distinguished from individuals with the other disorders or from controls. Although brain structural abnormalities have been found in affected individuals, including reductions in cortical surface area and thickness and in the size of subcortical structures^{17–22}, these are heterogeneous and widespread. Across the three disorders, the effect sizes for structural differences are generally in the same direction and scale in the order of those for schizophrenia being greater than those for bipolar disorder, which are in turn greater than those for MDD²³. Some of the measures, in particular cortical thickness, seem to be influenced by the severity and duration of illness and by pharmacological treatment²³.

Heritability from genetic epidemiology

Heritability estimates for schizophrenia, bipolar disorder or MDD (Table 1) derived from twins ascertained in hospital settings are generally higher than those obtained from population studies. These differences might reflect greater diagnostic homogeneity in hospital-based

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twin studies and/or the inclusion of more severe cases with a stronger genetic component. However, the lower estimates of heritability obtained from population studies probably align more closely to the samples used for genomic studies²⁴ and thus may be more relevant to discussions of the expected heritability from genetic epidemiology

that is not explained by current genomic data. The finding that heritability estimates are less than 100% is also consistent with a role for environmental factors, such as in utero or obstetric events that affect brain development and adverse childhood experiences such as abuse or neglect^{12–14}. Finally, genetic epidemiology has provided evidence

Fig. 1 | Summary of genomic findings for schizophrenia, bipolar disorder and depression. **a**, The effect sizes and frequencies of known risk variants for schizophrenia, bipolar disorder and depression. The effect sizes (odds ratios (OR)) are plotted against minor allele frequencies in the general population for copy number variants (CNVs), damaging rare coding variants (RCVs) (protein-truncating variants (PTVs) or missense variants) and common single-nucleotide polymorphisms (SNPs). The y-axis spacing is on a log₁₀ scale. The CNV and RCV effect sizes are from studies of schizophrenia^{39,48}. For most CNVs, the cytogenetic location is given, where ‘dup’ and ‘del’ refer to duplication and deletion, respectively. The CNV loci that have been attributed to named syndromes are indicated: WBS denotes the region deleted in Williams–Beuren syndrome and PWS/AS denotes the region associated with Prader–Willi syndrome and Angelman syndrome. RCV-associated genes are derived from refs. 48,49. *AKAP11* was also identified in a meta-analysis of bipolar disorder and schizophrenia datasets⁵¹. The RCV effect sizes refer to the excess burden of RCVs in the named gene. CNV and RCV effect sizes are imprecise owing to the small number of observations. The genome-wide SNP association data for schizophrenia, bipolar disorder and depression are derived from refs. 31,33,36 (also represented in part **b**). The effect sizes for polygenic risk scores (PRSs) for schizophrenia, bipolar disorder and depression are plotted on the basis of the largest test samples

(those of European ancestries). The PRS data for bipolar disorder are from ref. 29, as equivalent data were not reported in the more recent publication³⁶. The abbreviations 10 or 100 M denote comparisons between the top decile (10 M) or top centile (100 M) of the PRS and the median decile. The abbreviation 100 B denotes the comparison between the top and bottom PRS centiles (data for bipolar disorder not available). The insert below the main plot depicts the common variant associations for each of the disorders at higher resolution to visualize the effect sizes. Part **a** is adapted from ref. 7, CC BY 4.0 (<https://creativecommons.org/licenses/by/4.0>), adding the data for bipolar disorder and depression. **b**, Genome-wide association study (GWAS) sample sizes, ancestries and associated loci. The pie charts illustrate the proportions of cases denoted as of European, East Asian, South Asian, African or African–American, Latino/Hispanic or multiple/mixed background in the largest published GWAS of schizophrenia, depression and bipolar disorder. The first three charts (from left to right) show the largest available GWAS data for schizophrenia³¹, depression³³ and bipolar disorder³⁶. The last two charts show the largest studies of cases described as meeting clinical diagnostic criteria for major depressive disorder (MDD)¹⁴⁶ and bipolar disorder²⁹, demonstrating the relatively low yield of associations per clinically defined case, particularly for depression, which has motivated the inclusion of minimal-phenotype cases in the largest studies.

that these disorders are not completely distinct genetically, with both shared and unique genetic risk factors^{25–27}.

Genetic architectures of schizophrenia, bipolar disorder and depression

Here, we review genomic studies of schizophrenia, bipolar disorder and depression, structured around the different classes of genetic variants that have been studied in greatest depth. We also consider how differences in depth of phenotyping can affect the results of genomic studies as well as the gap between heritability explained by genomic studies and that expected from genetic epidemiology.

Common variants

Schizophrenia, bipolar disorder and MDD are among the most polygenic of human disorders²⁸; around 10,000 or more common variants are estimated to contribute to each, with MDD being the most polygenic^{29,30}. Large-scale GWAS have identified hundreds of common risk loci for these disorders at genome-wide significance (Fig. 1a), with the most significantly associated variants at each locus typically residing in non-coding regions. Of note, an important limitation of GWAS in psychiatry, as in other disorders, has been the disproportionate dependence on participants of European ancestries (Fig. 1b).

Schizophrenia. The largest GWAS of schizophrenia identified 287 distinct genomic loci that are associated with this disorder³¹, many of which contain multiple independent association signals (Fig. 1b). Fine mapping integrated with functional genomic data prioritized 120 genes as candidates for mediating associations. Only 16 of these genes contained a credible nonsynonymous or untranslated region variant, with the strongest evidence (posterior probability >0.99) for a causal missense variant in *SLC39A8* and a 3′ untranslated region variant in *FURIN*. A key finding was that fine-mapped associations with common variants were enriched for genes that were also implicated by rare coding variants (RCVs) in schizophrenia (see ‘RCVs’ section), as well as in autism and intellectual disability.

Depression. GWAS of clinical cases of depression have had limited success, with the largest such study identifying seven risk loci (Fig. 1b),

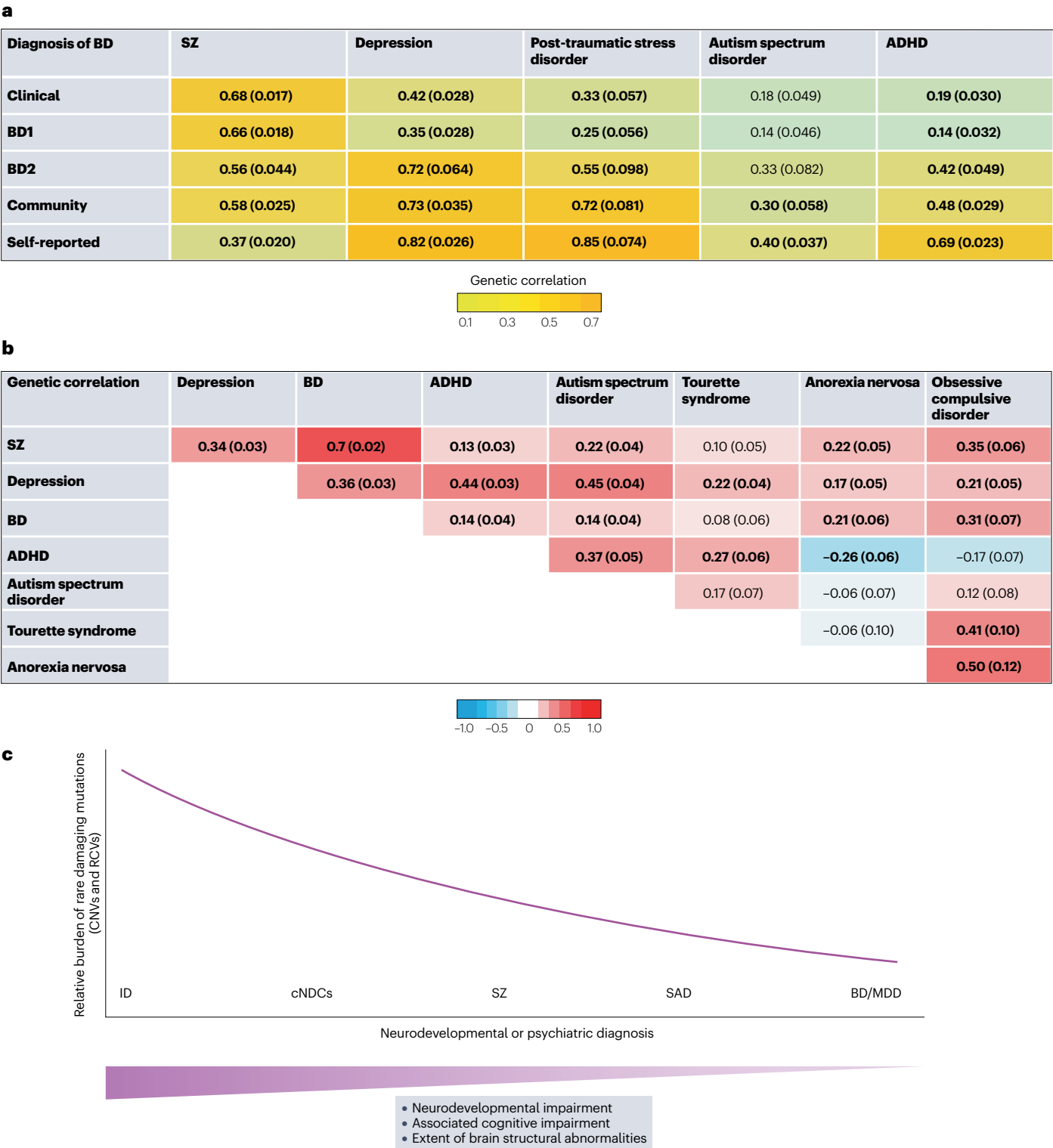
owing to the power challenge of studying a disorder of modest heritability but high prevalence³² (Table 1). The need for larger studies has been met by including cases who were diagnosed on the basis of minimal phenotyping, many of whom may not meet diagnostic criteria for MDD (Box 1 and Fig. 1b). We use the term depression, rather than MDD, to describe the phenotype of such studies.

The ‘minimal-phenotype’ approach has been highly effective for identifying risk variants for depression. However, although the results of GWAS indicate that research-based clinical diagnoses and self-reported diagnoses have high levels of genetic correlation (r_g 0.78–0.88)³³, the minimal-phenotype approach may fail to detect alleles that are specific for MDD while enriching for those that are shared with other psychiatric and behavioural traits^{34,35}. The largest published GWAS of depression to have been published so far, in which most cases were defined by minimal phenotyping or through electronic health records, identified 635 risk loci³³ (Fig. 1b). Gene prioritization highlighted 308 genes as candidates for mediating the associations, 21 of which were also prioritized in the GWAS of schizophrenia³¹, including *SP4*, *FURIN* and *CACNA1C*.

Another key observation from this study³³ was that polygenic risk scores (PRSs) derived from this sample of predominantly European ancestries performed poorly at indexing liability in people of non-European ancestry. In Europeans, the PRS methodology indexed around 4% of liability, whereas the corresponding figures in populations of Asian and African ancestries (including African–American populations) were around 1% and 0.2%, respectively.

Bipolar disorder. The largest GWAS of bipolar disorder identified 298 associated loci³⁶, powered mainly from self-reported cases (which comprised 58% of the sample). Compared with cases with a research-based clinical diagnosis, self-reported bipolar disorder had lower values for single-nucleotide polymorphism heritability (0.08 compared with 0.21) and stronger genetic correlations with a broad range of other psychiatric disorders (Fig. 2a), the latter implying that the increased power from the inclusion of self-reported cases comes at the cost of some loss of specificity for the clinical disorder. Gene prioritization incorporating fine-mapping and multiple functional genomics annotations highlighted 36 genes as candidates

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for mediating the associations, 15 of which were also prioritized in the depression GWAS³³ and 3 in the schizophrenia GWAS³¹. Notable among these are *SP4* and *FURIN*, which have been prioritized in schizophrenia, depression and bipolar disorder^{31,33,36}. Reminiscent

of the findings in schizophrenia³¹, the fine-mapped genes identified as mediating associations were also enriched for deleterious RCVs in exome-sequencing studies of bipolar disorder, as well as for deleterious RCVs in schizophrenia.

Fig. 2 | Genetic correlations and impact of minimal phenotyping, and the neurodevelopmental continuum and gradient. a, The effect of diagnostic method on genetic correlations between bipolar disorder and other psychiatric conditions. The numbers shown are for genetic correlations between pairs of disorders (with standard errors in brackets). The bold font indicates that the correlation is statistically significant after Bonferroni correction. Bipolar disorder diagnoses, in decreasing order of stringency, are clinical (based on structured or semistructured clinical interviews), community (based on data extracted from medical records, registries and questionnaires) and self-reported (based on self-reported surveys). Diagnoses of bipolar disorder type 1 (BD1) and bipolar disorder type 2 (BD2) were made from clinical and community datasets. The coloured shading denotes the extent of the correlation, from high positive correlation (orange) to low positive correlation (yellow). The data are derived from ref. 36. **b,** The genetic correlations between psychiatric

diagnoses. The numbers shown are for genetic correlations between pairs of disorders (with standard errors in brackets). The bold font indicates that the correlation is statistically significant after Bonferroni correction. The coloured shading denotes the scale and direction of the correlation, from positive (red) to negative (blue). The data are derived from ref. 107. **c,** The neurodevelopmental continuum and gradient. This graph illustrates the relationship between the relative burden of rare damaging mutations (copy number variants (CNVs) and rare coding variants (RCVs)) and neurodevelopmental and psychiatric diagnoses. The model postulates a gradient of neurodevelopmental impairment, with associated cognitive impairment and structural brain abnormalities^{15,16}. ID, intellectual disability; cNDCs, childhood-onset neurodevelopmental conditions (such as autism and attention deficit hyperactivity disorder (ADHD)); SZ, schizophrenia; SAD, schizoaffective disorder; BD, bipolar disorder; MDD, major depressive disorder.

Rare CNVs

Schizophrenia. Recurrent 22q11.2 deletions were the first genomic variants to be unequivocally associated with schizophrenia³⁷. There is now strong evidence for up to 13 schizophrenia-associated, recurrent, large (>20 kb), rare (frequency <0.01) CNVs^{38,39} and that the reciprocal duplication at the 22q11.2 locus is protective for the disorder^{39–41}. Apart from deletions within *NRXN1*, all of the CNVs that are robustly associated with schizophrenia are multigenic. Specific CNVs associated with schizophrenia have a frequency in cases of 0.01–0.6% and have relatively moderate-to-large effect sizes (Fig. 1a). Collectively, around 2% of cases of schizophrenia carry a known associated CNV, but discovery is incomplete^{38,39}. As discussed later, CNVs that are implicated in schizophrenia have also been implicated in a range of NDCs.

Bipolar disorder. CNVs seem to have a less important role in bipolar disorder than in schizophrenia. The largest studies of de novo (>100 kb)⁴² or case–control⁴³ CNVs found no excess of CNVs in cases of bipolar disorder. Moreover, the latter study found no evidence for enrichment in bipolar disorder of CNVs implicated in schizophrenia or NDCs, although there was some evidence for enrichment of these CNVs in a small sample of cases with schizoaffective disorder (Box 1).

Depression. NDC-associated CNVs are enriched in individuals with self-reported depression in the UK Biobank⁴⁴ and in those with MDD diagnoses derived from electronic records in a US multi-ancestry biobank⁴⁵. In the UK Biobank, those individuals with the most conservative definition of depression had the strongest enrichment for these CNVs (odds ratio (OR) 1.51), which was lower than for the same set of CNVs in schizophrenia (OR 2.56)³⁸. Three of the NDC-associated CNVs (1q21.1 duplication, Prader–Willi syndrome duplication and 16p11.2 duplication) reached experiment-wide significance for depression, although larger studies are required to confirm these results at stringent levels.

RCVs

Schizophrenia. The increased de novo mutation rate⁴⁶ and mutation burden⁴⁷ in cases have established the involvement of RCVs in schizophrenia. In the largest exome-wide study, that of the Schizophrenia Exome Meta-Analysis (SCHEMA) consortium⁴⁸, cases of schizophrenia were enriched for ultrarare (minor allele frequency <0.00002), damaging missense variants and protein-truncating variants (PTVs) in loss-of-function mutation-intolerant genes, particularly in genes expressed in neurons and at neuronal synapses, in genes with evidence for association in the recent schizophrenia GWAS³¹ and in genes implicated by RCVs in NDCs. More importantly (Fig. 1a), ten genes were enriched at

exome-wide significance for RCVs, two of which (*GRIN2A* and *SP4*) were also prioritized by fine mapping in the recent schizophrenia GWAS³¹. A subsequent meta-analysis of sequencing data from 161 genes and the SCHEMA dataset identified two additional risk genes, *AKAP11* and *SRRM2*, for schizophrenia at exome-wide significance⁴⁹, but many more remain to be discovered⁴⁸.

Around 1% of cases of schizophrenia carry a qualifying RCV in an associated gene⁴⁸. The rarity of the implicated mutations, with a frequency typically of less than 0.00002, precludes estimation of allelic effect sizes, but at the gene-burden level, point-estimate ORs for RCVs in schizophrenia are similar to those for CNVs (Fig. 1a).

Bipolar disorder. Burden tests suggest that rare PTVs in loss-of-function mutation-intolerant genes contribute to bipolar disorder⁵⁰ but less so than for schizophrenia (bipolar disorder, OR 1.11; schizophrenia, OR 1.26)⁵¹. Although the largest exome-sequencing study of bipolar disorder⁵¹ did not identify any associated genes, it did find that PTVs within genes implicated in schizophrenia were enriched in cases of bipolar disorder. Furthermore, a single gene, *AKAP11*, which was subsequently implicated at exome-wide significance in schizophrenia alone⁴⁹, was identified in this study at exome-wide significance by a combined schizophrenia–bipolar disorder analysis.

Depression. Evidence for RCVs in depression comes from a UK Biobank study⁵² in which cases defined according to electronic health records were enriched for PTVs (OR 1.17), at a slightly higher rate than in bipolar disorder but less so than in schizophrenia. Cases of depression were also enriched (OR 1.08) for damaging, missense RCVs. RCVs in cases of depression were enriched in genes that were also implicated by RCVs in schizophrenia, developmental delay and intellectual disability and autism and in genes with relatively high specificity for brain expression. That specificity was not seen for RCVs identified in cases with broader definitions of depression.

The heritability gap

Only a fraction of the heritability of schizophrenia, bipolar disorder and depression that is expected from genetic epidemiology is currently explained by common variants, CNVs and RCVs (Table 1). Some of this heritability gap might be explained by diagnostic misclassification and/or high levels of heterogeneity. This explanation is consistent with a study in which most cases met operational criteria for MDD rather than broader-based depression diagnoses, which found a much higher single-nucleotide polymorphism heritability of 0.24 (standard error 0.03)⁵³ than was observed in the largest meta-analyses of depression

discussed above (0.084, standard error 0.0007) (Table 1). Similar conclusions can be inferred from a recent study of bipolar disorder³⁶. Other explanations for the heritability gap are not specific to psychiatric disorders and are extensively discussed elsewhere (for example, refs. 54,55).

Pleiotropy of schizophrenia, bipolar disorder and depression

Family studies provide evidence for familial coaggregation between schizophrenia, bipolar disorder and MDD, with genetic correlation estimates of 0.30–0.7 (refs. 24,26,56–58). Genomic studies are now beginning to confirm that, at least in part, these correlations reflect pleiotropy. These findings also point to a lack of pathogenic validity of psychiatric diagnoses, which are syndromic and lack diagnostic tests and biomarkers (Box 1).

Pleiotropy of common variants

Common variant liabilities for many psychiatric diagnoses are correlated⁵, which suggests that there is widespread pleiotropy (Fig. 2b). There is a substantial genetic correlation between schizophrenia and bipolar disorder and moderate genetic correlation between depression and both schizophrenia and bipolar disorder. Consistent with there being genetic heterogeneity within bipolar disorder, common variant liability for bipolar disorder type 1 is strongly correlated with that for schizophrenia but only weakly correlated with that for depression, whereas the converse is true for bipolar disorder type 2 (Fig. 2a).

The extensive genetic correlations between schizophrenia, bipolar disorder and depression imply the existence of widespread allelic pleiotropy (shared risk alleles), but there is evidence that the level of genic pleiotropy (shared risk genes) may be even greater if we ignore allelic directions of effect²⁹. Thus, it has been estimated that almost all variant sites that influence liability to bipolar disorder also do so for schizophrenia, and around 90% of bipolar disorder variant sites are also shared with depression. Moreover, it was also estimated that for approximately 80% of the shared variant sites, the same alleles that were associated with bipolar disorder, were also associated with schizophrenia, and the same was true for the variant sites shared between bipolar disorder and depression⁵⁹. These findings imply that the biological processes underlying these disorders are more closely related than is evident from genetic correlations, although coaction with the unshared (or allelic discordant) variants and environmental exposures may mean that even the shared effects may be exerted in different brain regions, cell types and/or developmental stages in different disorders.

A caveat to the implication of widespread pleiotropy is that genetic correlation estimates can be inflated by assortative cross-disorder mating (where people with one disorder tend to pair with people with one of the other disorders) and by reciprocal diagnostic misclassification (where each discrete disorder is mistaken for the other)^{60,61} (Fig. 2a). It is unlikely that these factors fully account for the genetic correlations observed, particularly the high level of correlation between schizophrenia and bipolar disorder, which would require unrealistically high rates of reciprocal misdiagnosis^{61,62}. Moreover, misclassification can also deflate genetic correlations if additional cases are incorrectly assigned to one of a pair of the disorders in question³⁶. For example, there is a progressive deflation in genetic correlation between bipolar disorder and schizophrenia as the diagnostic stringency of bipolar disorder decreases from a clinical research-based diagnosis to diagnoses based on minimal (community) and low-depth (self-reported) phenotyping; in parallel, there is a substantial increase in genetic correlation between bipolar

disorder and other psychiatric conditions (Fig. 2a). These, and broadly similar findings from studies of depression⁶³, unsurprisingly suggest that low-depth phenotyping results in lower diagnostic specificity. Investigations aiming to distinguish alleles that are truly shared between and, perhaps more interestingly, specific to individual disorders will require deep phenotyping of large numbers of participants.

The question of whether genetic correlations are better accounted for by misdiagnosis of distinct disorders than by pleiotropy also needs to consider the fact that, as well as lacking biological validity, psychiatric diagnoses are heterogeneous and lack clear diagnostic boundaries (Box 1). Put simply, if the case for there being distinct diagnostic categories is weak, then this undermines the argument that genetic correlations reflect misdiagnosis rather than pleiotropy. If, as seems increasingly likely (see next paragraph), the clinical picture expressed by an individual is the result of the convergence of transdiagnostic groups or dimensions of symptoms and their partly independent underlying genetic risk factors, genetic correlation could be a consequence of genetic factors driving clinical similarity rather than reflecting the misdiagnosis of genetically distinct disorders. If this is true, the substantial overlaps in biological pathways that are implicated in these disorders (see 'Insights into the biology of schizophrenia, bipolar disorder and depression' section) may point to the biological basis of particular clinical features better than they do to the clinical diagnoses.

Studies aiming to use genomics for diagnostic stratification have shown that genetic liability for schizophrenia is greatest in people with schizophrenia, followed by those with schizoaffective disorder–bipolar subtype, then bipolar disorder type 1 followed by bipolar disorder type 2, and then unaffected individuals^{64–66}. Moreover, in people with bipolar disorder, schizophrenia liability is greatest in those with psychotic symptoms and greater still in those with mood-incongruent psychotic symptoms, which are the symptoms most similar to schizophrenia^{64,65}. Conversely, in those with schizophrenia, genetic liability for bipolar disorder is associated with the level of manic symptoms⁶⁵ and liability for MDD is associated with depressive symptoms⁶⁷. Finally, there is evidence that in people with bipolar disorder, the associations between schizophrenia PRS and mood-incongruent psychosis do not derive from shared schizophrenia and bipolar disorder alleles but from a component of liability that is fairly specific to schizophrenia⁶⁸. Thus, pleiotropy not only points to the genetic relationships between psychiatric diagnoses but also begins to offer explanations for heterogeneity in transdiagnostic clinical features within diagnostic categories. In addition, the PRSs for both schizophrenia and depression are associated with worse treatment response and worse outcomes in those with bipolar disorder^{69–71}.

Pleiotropy of rare variants

The same rare CNVs and the same RCV-containing genes that are associated with schizophrenia are also enriched in depression and bipolar disorder, respectively (see 'Genetic architectures of schizophrenia, bipolar disorder and depression' section). These CNVs and RCV-associated genes are also often pleiotropic for NDCs^{38,48,72–76}, and moreover, the specific mutations that confer risk of NDCs⁷⁷ are enriched in people with schizophrenia, which implies that there is a degree of overlap in the primary molecular pathology.

The overall contribution of CNVs and RCVs to NDCs, schizophrenia, bipolar disorder and MDD is not uniform, with their enrichment being greatest in disorders with the strongest neurodevelopmental contribution (as inferred from onset in childhood and greater cognitive impairment) (Fig. 2c). Thus, intellectual disability is most strongly enriched for

CNVs and RCVs, then autism and ADHD, then schizophrenia¹⁶, with lower contributions of CNVs and RCVs to both MDD and bipolar disorder. There is also evidence that the genes and mutation sites are most highly conserved in intellectual disability, then autism and then schizophrenia⁴⁶. For some genes, different classes of rare mutation might be implicated in NDCs and schizophrenia⁴⁸, but there is as of now no evidence relating to the relative severity of the different classes of mutation. Within schizophrenia, autism and ADHD, the burden of rare mutations is also not uniform, being highest in those with severe cognitive impairment, although it is still increased in those without cognitive impairment^{76,78,79}.

Common variants modulate effects of rare variants

Evidence from studies of schizophrenia^{80–82}, depression⁵² and NDCs⁸³ indicates that specific outcomes in carriers of rare variants are, at least in part, a consequence of additive coaction with common variants that underlie liability to these disorders. These findings suggest that rare and common variants confer overall risk together in a manner consistent with a polygenic liability threshold model, rather than that the two classes of variant (rare and common) lead to distinct, heterogeneous outcomes⁸¹. This also implies that PRSs and, indeed, probably other risk factors – both genetic and environmental – could be used for risk stratification in individuals with rare, large-effect but incompletely penetrant risk alleles⁸¹.

Insights into the biology of schizophrenia, bipolar disorder and depression

Here, we review the main findings from studies that have applied biological annotations to genomic data for schizophrenia, bipolar disorder and depression to elucidate their underlying neural bases. We also describe emerging technologies for further investigating the molecular and cellular risk mechanisms for these disorders.

Neuroanatomical and cellular insights

Analyses of genomic data in relation to brain region-specific and cell-specific gene expression patterns have provided insights into the likely neuroanatomical and cellular mediators of genetic liability to psychiatric disorders (Fig. 3). Consistent with neuroimaging data^{17–22}, common variants associated with these conditions are strongly enriched in genes with high expression specificity for the cerebral cortex, as well as subcortical brain regions involved in cognitive and affective processes, including the hippocampus, amygdala and striatum^{29,31,84}. In schizophrenia, the pattern of RCV enrichment according to the regional expression patterns of genes in the brain mirrors that of common variant enrichment⁴⁸. RCV studies of bipolar disorder and MDD are underpowered, but a small enrichment of RCVs in genes expressed in the amygdala has been reported for bipolar disorder⁵¹.

Analyses of common risk variant enrichment in predicted regulatory genomic elements and in genes with high expression specificity for individual cell populations have shown that genetic liability for schizophrenia, bipolar disorder and MDD operates mainly through neurons (rather than glia) in the postnatal brain^{85,86}. The most strongly implicated neuronal populations are glutamatergic neurons of the cerebral cortex, hippocampus and amygdala, cortical GABAergic interneurons and GABAergic medium spiny neurons of the striatum^{29,31,85–87}. Common risk alleles for schizophrenia, bipolar disorder and MDD are also enriched in putative regulatory genomic elements identified in the human prenatal brain^{88–90}. However, there is conflicting evidence as to the prenatal cell types that mediate common variant liability to these conditions; some studies have reported pronounced enrichment

of common risk variants in regulatory elements operating in neural progenitor cells^{89,91}, whereas other studies (based on cell-specific gene expression and/or regulation) have reported enrichments of common risk variants in developing, postmitotic neuronal populations^{92–96}.

Although there is considerable overlap in the cell types implicated in each of the psychiatric disorders, there is evidence that certain cell populations may have a more important role in individual diagnoses. For example, a recent study found enrichment of common variants associated with depression but not schizophrenia or bipolar disorder in regulatory genomic regions mapped in GABAergic neurons of the foetal midbrain⁹⁷. Larger single-cell datasets and the incorporation of spatial transcriptomics (see ‘Emerging tools for biological insights’ section) should help to further resolve the cell populations that are most important for mediating genetic risk for these conditions.

Insights into biological processes

Ontology analyses of GWAS findings for schizophrenia, bipolar disorder and depression are consistent with these disorders being associated with neuronal disturbances, with enrichment of common risk variants in genes involved in synaptic function, ion channel activity and neuronal development^{29,31,98}. The CNVs identified in schizophrenia are also enriched for synaptic genes^{39,99,100}, and the RCVs identified in schizophrenia are enriched in genes associated with synaptic function, ion channel activity and chromatin modification⁴⁸. The observations that genes related to chromatin modification are also enriched for RCVs in NDCs¹⁰¹ but are not notably enriched for common variant associations in schizophrenia³¹ may partly explain the particular role postulated for rare variation in the neurodevelopmental pathology of schizophrenia¹⁶.

Individual genes implicated in the pathobiology of schizophrenia that have strong support from both GWAS and RCV analyses include *GRIN2A*, which encodes a subunit of the glutamatergic NMDA receptor, and *SP4*, which encodes a transcription factor known to regulate NMDA receptor subunit expression¹⁰². Together, these results substantiate the long-hypothesized role of the NMDA receptor in schizophrenia pathophysiology^{103,104}. Common variants in several genes with roles in neuronal development and function (such as *FURIN*, *SP4*, *DCC*, *CACNA1C* and *ZNF804A*) seem to confer risk for bipolar disorder, depression and schizophrenia^{29,31,33,98}. However, there is evidence that variation at other genes confers risk for specific disorders. For example, schizophrenia-associated variation affecting the expression of *C4A*¹⁰⁵ does not seem to be associated with depression¹⁰⁶ or bipolar disorder²⁹, and discordant risk alleles for schizophrenia and depression at a locus on chromosome 5 (refs. 107,108) are associated with increased or decreased expression, respectively, of genes in the protocadherin alpha cluster^{108,109}.

Emerging tools

The translation of genetic findings into an improved understanding of molecular and cellular risk mechanisms will benefit from advances in functional genomics and cell and organoid culture technologies. For example, information from spatial ‘omics’ assays (for example, ref. 110) – by which global gene expression and epigenomic features can be measured in individual cells in their native tissue context – will help to determine the precise location of cells mediating genetic risk for these conditions (for example, ref. 111). Single-cell RNA-sequencing can be combined with genotype data to identify expression quantitative trait loci operating in individual brain cell populations, which, when integrated with GWAS data, can shed light on the cell-specific molecular risk mechanisms through which common, non-coding risk variants might operate¹¹². Advances in CRISPR technology (such as CRISPR

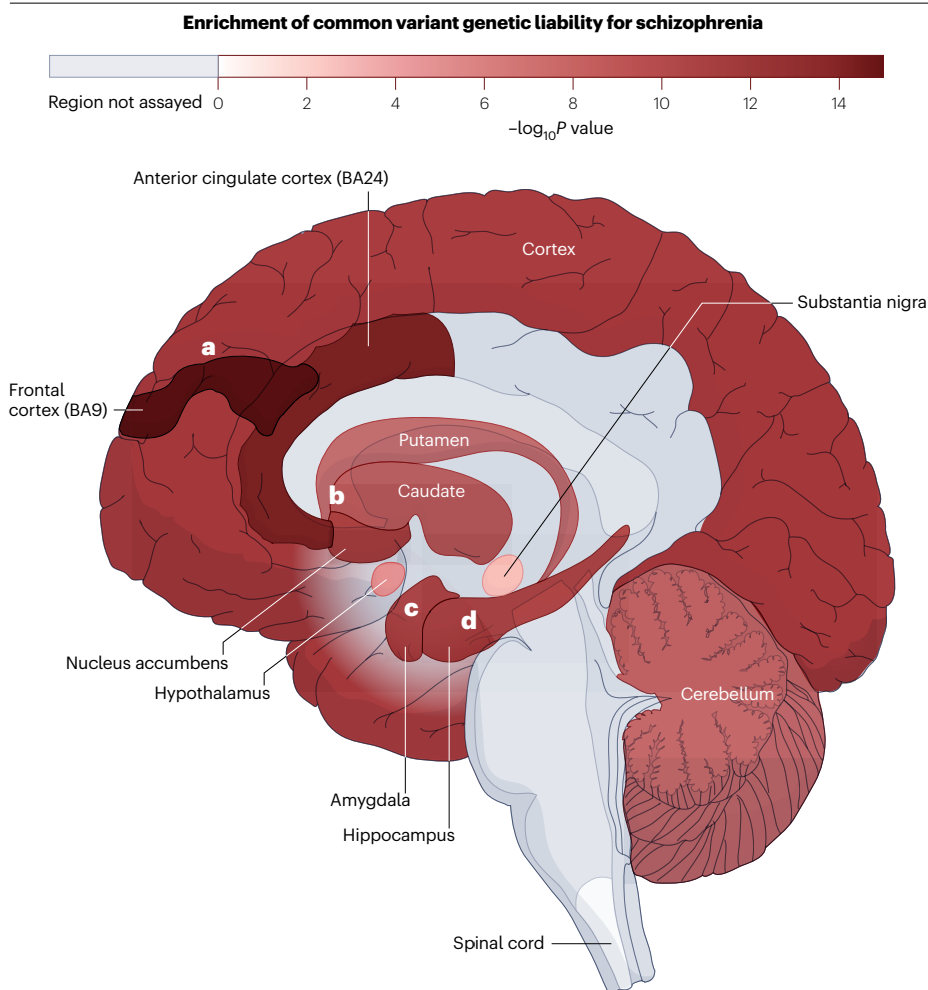
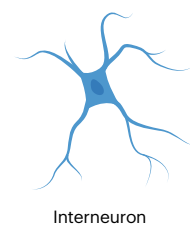
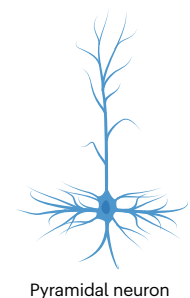


Fig. 3 | Neuroanatomical and cellular enrichment of common variant liability to schizophrenia based on gene expression specificity. The neuroanatomical heat map shows the mean $-\log_{10}(P$ value) for enrichment of common variant genetic liability for schizophrenia, as assessed by MAGMA¹⁴⁷ and S-LDSR¹⁴⁸ in ref. 31, in genes with high RNA expression specificity for each of eleven regions of the adult human central nervous system assayed by the GTEx Consortium¹⁴⁹: frontal cortex, cortex and anterior cingulate cortex (collectively, the 'cerebral cortex'); putamen, caudate and nucleus accumbens (collectively, the 'striatum'); hypothalamus; amygdala; hippocampus; substantia nigra; spinal cord (cervical C1); and cerebellum. The most significant enrichments for genetic liability are observed in genes with high expression specificity for assayed regions of the cerebral cortex, whereas the least significant enrichments for genetic liability are observed in genes with preferential expression in the substantia nigra and spinal cord (cervical C1). The GTEx Consortium did not provide RNA assays for non-coloured brain regions, and therefore, no conclusions can be drawn as to their enrichment for genetic liability. The genes with high expression specificity for glutamatergic pyramidal neurons and GABAergic interneurons of the cerebral cortex, glutamatergic granule cells and pyramidal neurons of the hippocampus, glutamatergic pyramidal neurons of the amygdala and GABAergic medium spiny neurons of the striatum are prominently enriched for schizophrenia-associated common variants^{31,86,87}. Similar patterns of neuroanatomical and cellular enrichment of common variant liability have also been reported for bipolar disorder²⁹ and major depressive disorder (MDD)⁸⁴. BA9, Brodmann area 9; BA24, Brodmann area 24.

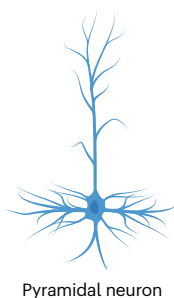
a Cerebral cortex



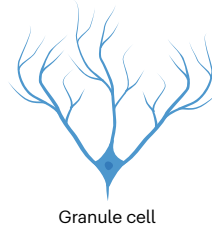
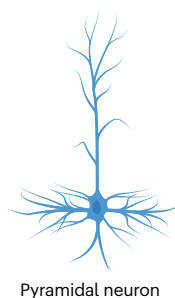
b Striatum



c Amygdala



d Hippocampus



activation and CRISPR interference of gene transcription) can be used to model the effects of common, non-coding risk variants (as well as RCVs) on gene regulation in relevant human neural cell types or multicellular brain organoids *in vitro*¹¹³, with single-cell RNA-sequencing providing an efficient means of assaying in parallel the effects of multiple gene perturbations on cellular gene expression¹¹⁴. Given the extensive use of alternative exons in the human brain transcriptome¹¹⁵, it is important that such modelling takes into account the specific RNA transcripts of each gene that are affected by genetic risk variation, as well as the developmental timing of these effects.

Clinical implications for schizophrenia, bipolar disorder and depression

The potential benefits, limitations and challenges of genomic testing in psychiatry have been reviewed elsewhere^{116–118}, and a detailed review of these complex issues is beyond the scope of this article. Here, we consider issues of immediate relevance to patients, clinicians, policymakers and the public: the implications of genomic findings for population screening, the potential utility of genomic data in clinical practice and the clinical opportunities offered by pharmacogenomics. Some of the challenges facing the application of genomics in psychiatry are considered in Box 2.

Population screening

Clearly, risk prediction for psychiatric disorders in the general population using genomic data alone will be limited by the fact that genetic factors account for only a proportion of the population variance in liability¹¹⁹. In addition, only a fraction of the genetic liability is currently explained by known genomic variation (Table 1). The specificity of population testing will also be limited by the low population prevalences of these psychiatric disorders, particularly of schizophrenia and bipolar disorder. In any case, given the lack of proven preventative, health-management interventions in psychiatry, the current utility of population screening is questionable, and, in some instances, this could be harmful¹¹⁸.

Diagnostic and prognostic potential of PRSs

Even though common variants currently account for only a proportion of the heritability of psychiatric disorders, it seems likely that, as genomic data accumulate, the use of PRSs – together with lifestyle information, clinical data and biomarkers – may improve patient stratification and outcome prediction and aid clinical decision making in symptomatic individuals who do not meet diagnostic criteria but where clinical signs and symptoms suggest that the risk of illness is higher than in the general population¹¹⁶. For example, in individuals with at-risk mental states, a combination of clinical information and PRS might identify those individuals for whom early treatment or closer follow-up is required¹¹⁶. However, high-quality, large-scale clinical studies are required to determine the clinical utility of PRSs and their integration with other data sources to identify the patient groups that will benefit and the outcomes that are best predicted. Moreover, the current unrepresentative nature of published genetic studies in relation to the global population, particularly with respect to ancestry, raises challenges as to the equitable application of PRSs in clinical contexts, as discussed above for the genetic architecture of depression.

Clinical testing for rare variants

Clinical testing for rare variants is available in many countries for children with NDCs. As data implicating rare variants and particular CNVs

in adult psychiatric disorders have accumulated, there have been calls for testing to be available at least in high-risk groups, such as those with premorbid cognitive impairment, a history of childhood-onset NDC, congenital malformations, dysmorphic features, developmental delay or a family history of schizophrenia or NDCs¹²⁰. The value of such testing is now recognized in several guidelines and is being implemented internationally¹²¹. Nonetheless, further research to clarify the benefits and harms of clinical testing, particularly beyond high-risk groups, is required, as is close integration with genetic counselling services^{117,122}. Moreover, given that many of the rare variants that confer risk for psychiatric disorders are also associated with NDCs as well as physical health conditions that may be treatable or preventable, appropriate cross-disciplinary assessment and care should be accessible¹²³.

Pharmacogenomics

Pharmacogenomics seeks to characterize the genetic contribution to interindividual variation in the response to drugs, both in terms of efficacy and safety¹²⁴. Perhaps the greatest potential for the clinical translation of pharmacogenomics in psychiatry is with respect to

Box 2 | Roadblocks to the clinical application of psychiatric genomics

Currently, the impact of genomics data on psychiatric care is limited, but it seems likely that this will change. Beyond the requirement for more research, the challenges for effective and ethical implementation of genomics analysis for psychiatric disease are not trivial and are different to those for other branches of medicine¹¹⁸. There is a need for improved genetics training to give psychiatrists and other members of the clinical team the knowledge and confidence to initiate testing and interpret the results¹¹⁸. Professional organizations will need to develop practice guidelines to ensure appropriate standards of care, and more extensive collaboration with other clinicians and disciplines will be required¹¹⁸. It will also be important for clinicians to consider issues around patient engagement, choice and informed consent, given the potential for harmful medical and social outcomes¹¹⁸. Finally, given the uptake of direct-to-consumer genetic testing and the greater public awareness of genomics in medicine, there is a need for greater genomic education for the general public as well as for clinicians.

More multi-ancestry studies and genome reference panels that reflect the ancestral diversity of human populations are essential to ensure that the clinical benefits of genomics are realized globally, as well as to obtain novel genetic and biological insights. Achieving those benefits will also require the development of research skills and research infra-structure in many places globally¹⁵⁴. There is a great deal to be done here, but this is starting to be addressed in psychiatry within the Psychiatric Genomics Consortium and through specific initiatives focusing on non-European ancestries, examples of which include the [Latin American Genomics Consortium](#), [Neuropsychiatric Genetics in African Populations](#) (NeuroGAP) and the affiliated Global Initiative for Neuropsychiatric Genetics Education in Research programme¹⁵⁵, the Ancestral Populations Network of the National Institute of Mental Health and [Depression Genetics in Africa](#).

Glossary

Burden test

Frequently used to increase the power of exome-wide association studies. They combine sets of variants within a gene to produce a single burden score that is then tested for association.

Common variants

Variants that are present in the population at frequencies of 1% or more. They include single-nucleotide polymorphisms, which can be assayed in parallel across the genome in genome-wide association studies.

Copy number variants

(CNVs). Variants for which the number of copies of a specific segment of DNA varies in the population. These structural differences between individuals may have arisen as a result of duplications, deletions or other changes.

Delusion

A fixed, false belief that is not amenable to change despite conflicting evidence.

Depression

A mental state that involves low mood and decreased activity.

Genetic correlation

(r_g). An estimate of the average similarity of causal allelic effects on two traits.

This usually refers to the genome-wide average, although it can be specified to be locus wide or chromosome wide. A high degree of genetic correlation is usually indicative of pleiotropy.

Hallucination

A perception in the absence of an external stimulus that has a compelling sense of reality.

Hypomania

A less severe form of mania that is not accompanied by marked functional impairments (for example, to occupation or personal relationships).

Loss-of-function mutation-intolerant genes

Genes with near-complete depletion of protein-truncating variants in population studies, suggesting that there is strong direct selection against such variants.

Mania

A mental state that involves elevated mood, which can be euphoric or irritable, and increased activity.

Mood-incongruent psychotic symptoms

Hallucinations or delusions that are not aligned with the person's current mood state.

Pleiotropy

Describes the phenomenon whereby a variant or gene influences more than one phenotype. The terms 'direct', 'horizontal' and 'biological' pleiotropy imply that the effects on the phenotypes are independent. The terms 'mediated', 'vertical' and 'indirect' pleiotropy imply that the effects on one phenotype are mediated through the other; for example, alleles that influence smoking behaviour are likely to have pleiotropic effects on all traits for which smoking is a risk factor.

Polygenic risk score

(PRS). Summarizes the estimated effect of many genetic variants on an individual's phenotype. It is typically calculated as a weighted sum of trait-associated alleles.

Protein-truncating variants

(PTVs). Variants that are predicted to shorten the coding sequence of a gene and which may result in reduced expression of that gene and its protein owing to nonsense-mediated decay. They include nonsense, frameshift and essential splice site variants.

Psychotic symptoms

Symptoms, particularly delusions and hallucinations, that involve difficulty in determining what is real from what is not.

Rare coding variants

(RCVs). Variants in the coding sequence of a gene that are present in less than 1% of the population. They can be detected by genome-wide sequencing studies.

pharmacokinetics, in particular, the genetic basis of drug metabolism. Accumulating evidence in recent years – including the identification of drug–gene pairs for psychotropic medications¹²⁵ – has led to expert recommendations for the testing of cytochrome P450 genes (*CYP2D6* and *CYP2C19*) to inform the selection and dosing of several commonly used antidepressant and antipsychotic medications, as well as for testing of human leukocyte antigen genes when using some mood stabilizers, including carbamazepine¹²⁶. These recommendations were supported by the findings of a large-scale study that carried out pre-emptive pharmacogenetic testing and showed that genotype-guided dosing reduced the incidence of adverse drug reactions, including to antidepressants and antipsychotics¹²⁷. The importance of considering ancestry to optimize antipsychotic dosing has been shown for clozapine¹²⁸. Furthermore, equitable access to clozapine, which is the only licensed medication for treatment-resistant schizophrenia, would be enhanced through the implementation of preinitiation genetic testing of the Duffy-null (*ACKRI*-null) genotype. The Duffy-null genotype is more common in individuals of African and Middle Eastern ancestries and is associated with lower neutrophil counts, so-called benign ethnic neutropenia. Benign ethnic neutropenia is not associated with clozapine-induced agranulocytosis but, if undiagnosed, may result in exclusion or cessation of clozapine treatment in patients of African and Middle Eastern ancestries owing to unfounded fears about the risk of agranulocytosis¹²⁹.

The genetic basis of treatment response has proven more challenging to delineate¹³⁰. Nonetheless, a genetic basis for variation in the response to lithium¹³¹ and antipsychotics¹³² has been demonstrated, although these findings are not currently clinically impactful.

Finally, although the above evidence supports the use of pharmacogenomic testing in some clinical settings, there remain challenges to its implementation within healthcare systems that need to be overcome if the potential of pharmacogenomics is to be realized¹³³.

Conclusions and future perspectives

Although much of the heritability of psychiatric disorders remains to be explained, genomics studies are beginning to characterize their genetic architectures. Schizophrenia, bipolar disorder and MDD are highly polygenic, with risk conferred by a combination of common alleles of small effect and rare alleles of large effect, although the relative contribution of the latter is greater in NDCs than in psychiatric disorders (Fig. 2c). Consistent with findings from structural brain imaging studies, common risk variants for these disorders are enriched in genes with preferential expression in cerebral cortex and subcortical regions having cognitive and affective functions. At the cellular level, common variant liability for these conditions is enriched in genes with high expression specificity for neurons, rather than glia, within these regions. Gene ontology analyses of implicated genes point to disturbances in neuronal development, electrophysiology and synaptic

function. Together, these findings are consistent with the hypothesis that schizophrenia, bipolar disorder and MDD are syndromes resulting from widely distributed neuronal pathology. Although the reported patterns of neuroanatomical and cellular enrichment of common variant liability in the three disorders are similar (Fig. 3), further work is required to understand the extent to which these patterns may differ between and within diagnostic groups and their relationships with individual clinical features.

Data from larger sequencing studies, exome sequencing, whole-genome sequencing and long-read sequencing can be expected to refine emerging biological insights and potentially explain more of the heritability of psychiatric disorders. Given current and emerging technologies, progress is likely to come initially from studies of the exome and of structural variation, but the reduction in the cost of whole-genome sequencing has made the study of non-coding, rare variants (ncRVs) tractable. Specific associations with ncRVs are beginning to emerge for traits such as height that are well represented in large biobanks¹³⁴, but studies of ncRVs in adult psychiatric disorders are small, and the findings are provisional and restricted to reports of excess burdens proximal to particular genomic features in people with schizophrenia¹³⁵. It seems inevitable that ncRVs contribute to adult psychiatric disorders, but there are important analytic challenges to demonstrating this. The very low frequencies of RCVs of large effect have meant that insights from exome sequencing have involved burden tests of RCVs that are predicted to be deleterious within the coding exons of genes, rather than tests of individual variants. Similar considerations will hold for ncRVs, but for now, there is no obvious genetic unit for burden testing of ncRVs and no robust methodology for predicting the subset of ncRVs that are highly likely to be highly deleterious. Developments in the functional annotation of the genome, linking regulatory elements to specific genes (for example, ref. 96), are likely to improve prospects in this regard.

Biological insights into psychiatric disorders will also depend upon better annotation of the brain transcriptome and proteome, with greater spatial and temporal resolution¹³⁶. Understanding the balance between neurodevelopmental impairments and direct effects on adult brain function across the different diagnostic groups – and the specific processes that are affected – are key knowledge gaps for future research.

Rare variant studies support the hypothesis that psychiatric disorders share pleiotropic neurodevelopmental abnormalities with NDCs, the severity of which varies across and within disorders and which is indexed by cognitive impairment and rare variant burden^{15,16} (Fig. 2c). These and other findings point to the potential importance of cognitive impairment and, more specifically, the extent to which cognitive function is worse than would be expected from the cognitive function of other family members, as a diagnostic stratifier in psychiatry, reflecting the involvement of neurodevelopmental abnormalities¹³⁷.

Common variant studies have revealed a high degree of genetic correlation between schizophrenia and bipolar disorder, with risk for both conditions being moderately correlated with risk for MDD. Early transdiagnostic studies of these disorders suggest that groups of related symptoms, such as psychotic, manic or depressive symptoms, have partly independent genetic risk factors that, in combination, influence the mix of symptoms experienced by an individual. In this sense, genomics data provide validation for stratification approaches based on symptom features within and across disorders. There is increasing interest in approaches to model the latent structure of the pleiotropic relationships between multiple psychiatric disorders, and these

models have identified broad factors that may underlie cross-disorder correlations^{68,107,138}. These cross-disorder findings, perhaps coupled with the transdiagnostic symptom research discussed above, have the potential to identify valid, biologically informed classes or dimensions that cross existing diagnostic categories and, thus, advance the classification of psychiatric disorders. The lack of consistency and availability of clinical phenotypic measures across disorders presents a challenge to these advances and, hence, deep phenotyping that is aligned transdiagnostically should be a priority for the field. Genomic studies of depression have shown the value of increasing sample sizes, regardless of the depth of phenotyping, for estimating the risk of MDD, but studies of cases that meet diagnostic criteria might better inform our understanding of the specific neurobiology of clinical forms of these disorders³⁵. Moreover, to identify biologically valid classes or dimensions of disease and to understand the genetic basis of heterogeneity, there is also the need for a greater focus on deeper phenotyping at the symptom level, including cognition and other biomarkers.

Despite substantial progress in psychiatric genomics, we are still some way from the goal of more effective and better targeted treatments. Challenges to identifying tractable biology are posed by the highly polygenic nature of psychiatric disorders, the complexity and inaccessibility of the brain, the involvement of neurodevelopmental as well as adult mechanisms, and the heterogeneity inherent in psychiatric diagnoses. On the positive side, there is potential for genomics studies to inform novel and more biologically relevant classifications of disease. Furthermore, the availability of in vitro cell and organoid models means that it is now possible to investigate the effects of current and emerging risk alleles, singly and in combination, on human neural function¹¹³. Progress will depend upon coordinated interactions between geneticists, neurobiologists and clinical researchers.

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